

METROPOLIS, IL, USA

WMO: 720170

Lat: 37.186N

Lon: 88.751W

Elev: 117

StdP: 99.93

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63851

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.0	-8.8	-18.6	0.7	-9.2	-15.3	1.0	-6.9	9.5	6.5	8.7	6.0	1.8	10	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	34.0	24.5	32.9	24.4	32.2	24.0	26.7	31.3	25.8	30.5	25.2	29.9	2.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.3	20.8	29.2	24.7	20.0	28.4	23.9	19.0	27.5	84.1	31.3	80.4	30.7	77.6	29.5	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.1	5.8	-16.6	35.8	5.4	1.8	-20.5	37.1	-23.6	38.1	-26.7	39.1	-30.6	40.4		
(5)			-16.7	28.0	5.1	1.6	-20.3	29.2	-23.3	30.1	-26.2	31.0	-29.8	32.2	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	1.9	3.8	9.1	14.6	19.9	24.2	25.7	24.7	21.4	15.0	8.4	4.1	(6)	
	DBStd	9.74	6.22	6.32	5.95	5.30	4.31	4.56	2.68	2.74	3.56	4.92	5.14	5.47	(7)	
	HDD10.0	862	260	192	91	18	0	6	0	0	0	11	88	197	(8)	
	HDD18.3	2272	509	408	291	130	37	13	0	1	12	129	299	442	(9)	
	CDD10.0	2489	9	17	62	158	306	433	487	455	342	165	40	14	(10)	
	CDD18.3	856	0	0	4	19	84	190	229	197	104	24	2	1	(11)	
	CDH23.3	8359	0	0	30	176	769	1940	2333	1862	995	247	7	0	(12)	
	CDH26.7	3283	0	0	2	23	236	804	1014	769	371	63	1	0	(13)	
(14) Wind	WSAvg	2.1	2.8	2.7	2.8	2.7	2.0	1.7	1.4	1.2	1.3	1.8	2.2	2.4	(14)	
(15) Precipitation	PrecAvg	1257	107	91	124	124	127	105	106	78	84	94	102	113	(15)	
	PrecMax	1849	197	224	281	407	280	260	252	234	259	273	248	236	(16)	
	PrecMin	739	31	30	48	9	11	21	36	10	0	3	18	32	(17)	
	PrecStd	241	41	43	57	69	60	56	60	48	59	61	59	56	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	21.8	26.4	28.9	32.4	35.2	36.8	36.1	34.0	31.2	24.0	20.5	(19)	
		MCWB	16.4	16.3	17.9	19.9	22.2	23.2	24.7	24.3	23.7	22.5	17.9	17.8	(20)	
	2%	DB	17.1	19.7	23.5	27.3	30.8	33.2	33.9	33.6	32.5	28.3	22.1	17.9	(21)	
		MCWB	14.7	15.2	16.7	18.2	21.7	24.0	25.7	23.8	23.3	20.7	16.4	15.5	(22)	
	5%	DB	14.9	17.0	21.3	25.5	29.4	32.4	32.8	32.4	30.9	26.2	19.7	15.6	(23)	
		MCWB	12.2	13.1	15.6	17.5	21.1	23.6	25.3	24.0	22.8	19.0	14.7	12.8	(24)	
	10%	DB	12.3	14.0	18.9	23.6	27.7	31.2	31.8	31.0	28.8	23.7	17.3	12.9	(25)	
		MCWB	9.6	11.0	13.8	16.9	20.2	23.2	24.8	23.7	21.7	18.0	13.1	10.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.3	17.9	19.3	21.9	24.1	27.2	27.8	27.6	25.4	23.7	20.1	18.1	(27)	
		MCDB	18.0	20.7	24.0	26.0	28.9	31.9	32.9	31.3	31.0	28.9	22.5	19.6	(28)	
	2%	WB	15.6	15.8	17.8	20.1	23.2	25.9	26.9	26.2	24.6	22.2	17.8	16.1	(29)	
		MCDB	16.7	18.2	22.1	24.8	28.2	30.5	31.8	30.7	30.0	26.2	20.1	17.7	(30)	
	5%	WB	12.7	14.0	16.4	19.0	22.4	25.1	26.2	25.4	23.9	20.8	15.6	13.7	(31)	
		MCDB	14.5	16.4	20.4	23.4	27.2	30.0	30.9	30.1	28.9	24.1	18.3	15.2	(32)	
	10%	WB	9.9	11.3	15.0	17.9	21.5	24.3	25.4	24.7	23.0	19.3	13.5	10.9	(33)	
		MCDB	12.0	13.7	18.3	22.0	25.9	29.1	29.9	29.2	27.1	22.6	16.4	12.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	9.4	10.7	12.2	11.2	11.6	10.5	11.2	12.4	12.4	11.1	8.7	(35)	
		MCDBR	11.7	12.8	13.7	13.9	13.0	12.1	11.4	12.7	14.1	14.9	14.0	11.4	(36)	
	5% WB	MCWBR	10.1	9.7	8.2	7.0	5.3	4.3	4.4	4.8	5.6	7.1	9.0	9.5	(37)	
		MCWBR	10.2	11.1	11.7	11.2	10.9	10.4	10.2	10.5	12.3	12.0	11.1	9.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.319	0.329	0.365	0.400	0.428	0.440	0.479	0.478	0.411	0.361	0.324	0.313	(40)	
		taud	2.459	2.413	2.370	2.274	2.250	2.228	2.145	2.136	2.321	2.444	2.491	2.481	(41)	
	Ebn at Noon		868	904	898	881	858	844	808	800	839	856	856	850	(42)	
		Edh at Noon	84	99	113	132	137	141	152	150	118	95	80	77	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.13	2.75	3.82	5.03	5.67	6.39	6.16	5.65	4.73	3.57	2.44	1.82	(44)	
	RadStd		0.21	0.34	0.34	0.32	0.48	0.40	0.38	0.29	0.29	0.39	0.26	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page